

CIR 2025/2531 — Qualified Electronic Ledger Mapping

eIDAS 2.0 Art. 45i compliance mapping for Goya Ledger.

Audit date: 2026-08-31 · Codebase: v0.13.3

| Self-assessment. No CAB has validated these claims.

Art. 45i requirements

1. Sequential ordering of data records

Requirement	Goya control	Evidence
Monotonic block height	<code>Block::height: u64,</code> <code>genesis = 0, each block =</code> <code>latest + 1</code>	<code>src/storage/</code> <code>traits.rs:19, src/</code> <code>mining.rs:71–76</code>
Parent hash chain	<code>Block::parent_hash:</code> <code>[u8; 32] — SHA-256 of</code> <code>predecessor</code>	<code>src/storage/</code> <code>traits.rs:21</code>
FIFO transaction ordering	<code>OrderingService</code> <code>batches endorsed txs from</code> <code>VecDeque</code>	<code>src/ordering/</code> <code>service.rs:11–13</code>
BFT round serialization	<code>VoteMessage::round:</code> <code>u64 — monotonically</code> <code>increasing</code>	<code>src/consensus/bft/</code> <code>types.rs:39</code>
Light client header chain	<code>HeaderChain::append()</code> <code>validates height</code> <code>monotonicity + parent</code> <code>linkage</code>	<code>src/light_client/</code> <code>header.rs:86–100</code>

2. Time stamping

Requirement	Goya control	Evidence
Block timestamp	<code>Block::timestamp: u64</code> <code>(UNIX UTC) on every block</code>	<code>src/storage/</code> <code>traits.rs:20</code>

Requirement	Goya control	Evidence
Transaction timestamp	Transaction::timestamp: u64 on every transaction	src/storage/traits.rs:98
RFC 3161 TSA	TsaProvider with monotonic serial, policy OID, accuracy metadata	src/tsa/mod.rs:98–107
DER-encoded timestamps	POST /api/v1/tsa/timestamp/der → application/timestamp-reply	src/api/handlers/tsa.rs:74–101
Timestamp verification	POST /api/v1/tsa/verify validates token signatures	src/api/handlers/tsa.rs:104–122
Trusted time source	trait TimeSource with System/NTP/Simulated, drift guard (5s default)	src/time_source.rs:50–70

3. Tamper evidence

Requirement	Goya control	Evidence
Hash chain	Block::parent_hash — SHA-256 back-link per block	src/storage/traits.rs:21
Merkle root	Block::merkle_root: [u8; 32] over transaction IDs	src/storage/traits.rs:22, src/mining.rs:109–110
Chain verification API	GET /api/v1/chain/verify — walks all blocks, reports first invalid height	src/api/handlers/chain.rs:27–78
Canonical block hash	compute_block_hash() = SHA-256(height parent_hash merkle_root timestamp proposer)	src/api/handlers/chain.rs:11–18
Audit log hash chain	AuditEntry::previous_hash + entry_hash — SHA-256 linked	src/audit.rs:134–139
Audit chain verification	verify_audit_chain() — walks log, returns first broken link	src/audit.rs:180–190
FIPS 140-3 KAT	run_hash_self_tests() — SHA-256 + SHA3-256 CAVP vectors at startup	src/crypto/haser.rs:59–93

4. Data origin authentication

Requirement	Goya control	Evidence
Block proposer signature	Block::signature + signature_algorithm (Ed25519 / ML-DSA-65)	src/storage/traits.rs:26–29

Requirement	Goya control	Evidence
Dual (hybrid) signatures	Block::secondary_signature — classical + PQC on same block	src/storage/traits.rs:38–42
Orderer signature	sign_block_with_provider() over SHA-256(height parent_hash merkle_root)	src/ordering/mod.rs:29–57
BFT vote signatures	VoteMessage::signature with domain GOYA-BFT-VOTE-V1	src/consensus/bft/types.rs:44–51
Quorum certificate	QuorumCertificate aggregates validated votes $\geq 2f+1$	src/consensus/bft/types.rs:106–116
Multi-org endorsements	Block::endorsements: Vec<Endorsement>	src/storage/traits.rs:32
eIDAS signature levels	SignatureLevel::{Simple, Advanced, Qualified} per Art. 3(10-12)	src/signature/mod.rs:41–59
Signing algorithm support	Ed25519, ML-DSA-65, SLH-DSA-128s, RSA, ECDSA-P256	src/identity/signing.rs:26–35
Validator key registry	ValidatorRegistry maps validator ID → ML-DSA-65 pubkey	src/consensus/bft/validator_registry.rs:16–17

5. Unique identifiers

Requirement	Goya control	Evidence
Transaction ID	Transaction::id — UUID v4	src/storage/traits.rs:95, src/block_creation.rs:21
Block height	Block::height: u64 — unique monotonic index	src/storage/traits.rs:19
Block hash	Deterministic SHA-256 from canonical fields	src/api/handlers/chain.rs:11–18
DID	did:goya:{pubkey_hex[..16]} — globally unique	src/identity/did.rs
TSA serial number	AtomicU64 monotonic per TsaProvider instance	src/tsa/mod.rs:100–101
Audit trace ID	UUID v4 per AuditEntry	src/audit.rs:129
Replay prevention	BlockStore::mark_tx_seen() / is_tx_seen() — dedup across restarts	src/storage/traits.rs:576–583

6. Immutability (append-only)

Requirement	Goya control	Evidence
		src/storage/traits.rs:516–1029

Requirement	Goya control	Evidence
No block deletion	trait BlockStore exposes write_block / read_block — no delete_block	
No audit deletion	trait AuditStore exposes append() — no delete()	src/audit.rs:231–243
Hash chain detection	Deletion/modification breaks parent_hash or previous_hash chain	Tests: mine_second_block_links_parent
7-year retention	DEFAULT_RETENTION_SECS = 7 years (ETSI TS 102 042)	src/audit_retention.rs:11–15
15-year UAE retention	UAE_RETENTION_SECS configurable per jurisdiction	src/audit_retention.rs

7. Consensus mechanism

Requirement	Goya control	Evidence
BFT protocol	HotStuff-inspired 4- phase: Prepare → PreCommit → Commit → Decide	src/consensus/bft/ types.rs:5–19
Fault tolerance	$f = (n-1)/3$, quorum = $2f+1$, minimum $n=4$	src/consensus/bft/ quorum.rs:50–79
Safety (no fork)	SafetyState with high_qc / locked_qc locking (two-chain/three- chain)	src/consensus/bft/ safety.rs:1–14
Formal safety proof	Lean 4: no_fork — two $2f+1$ quorums overlap by $\geq f+1$ honest	formal/BftSafety.lean
Leader rotation	Round-robin with exponential backoff timeouts	src/consensus/bft/ round_manager.rs:48–66
Equivocation detection	Dedicated module for conflicting-vote detection	src/consensus/ equivocation.rs
Slashing	Penalty mechanism for Byzantine validators	src/consensus/ slashing.rs
Ancestry verification	ChainAncestryChecker backed by block store, fail-closed	src/consensus/bft/ safety.rs:46–97

8. Auditability

Requirement	Goya control	Evidence
Semantic audit actions	40+ <code>AuditAction</code> variants (ETSI TS 102 042 aligned)	<code>src/audit.rs:19–107</code>
Hash-chained audit log	<code>AuditEntry::previous_hash</code> + <code>entry_hash</code>	<code>src/audit.rs:134–139</code>
Audit query API	<code>GET /api/v1/audit/requests</code> — paginated, filterable	<code>src/api/handlers/audit.rs:27–68</code>
CSV export	<code>GET /api/v1/audit/export</code>	<code>src/api/handlers/audit.rs:71–91</code>
Chain verification API	<code>GET /api/v1/chain/verify</code> + <code>GET /api/v1/chain/info</code>	<code>src/api/handlers/chain.rs</code>
Audit integrity check	<code>verify_audit_chain()</code> — cryptographic hash chain walk	<code>src/audit.rs:180–190</code>
Retention enforcement	Auto-purge disabled by default, configurable per jurisdiction	<code>src/audit_retention.rs</code>

Summary

Dimension	Status	Gaps
Sequential ordering	Covered	None
Time stamping	Covered	TSA not yet qualified (requires QTSP)
Tamper evidence	Covered	None
Data origin authentication	Covered	Qualified signatures require QSCD (Gap 10)
Unique identifiers	Covered	None
Immutability	Covered	None
Consensus	Covered	Formal proof exists (Lean 4)
Auditability	Covered	CAB validation pending

All 8 Art. 45i technical dimensions are implemented. Remaining blockers are organizational:

1. **Qualified status** requires QTSP certification (Gap 11) — the ledger itself is technically ready
2. **Qualified timestamps** require the TSA to operate under a QTSP
3. **Qualified signatures** require QSCD integration (Gap 10)
4. **CAB validation** of audit and security controls (Gap 6)